

Repetition Statements

- Loops are also known as Iterating statements or Looping constructs in Java.
- loops are used to repeat a block of code.
- In Java, there are three types of loops.
 - 1 while loop
 - 2 do...while loop
 - 3 for loop

Repetition Statements

while loop

- **Java while loop** is a control flow statement that allows code to be **executed repeatedly based on a given Boolean condition**.
- It **executes** the statements repeatedly until the **Boolean expression is true**
 - ▶ The **Boolean condition** becomes **false**, the loop automatically stops.
- The **while loop is considered as a repeating if statement**.

Repetition Statements

while loop

- Syntax

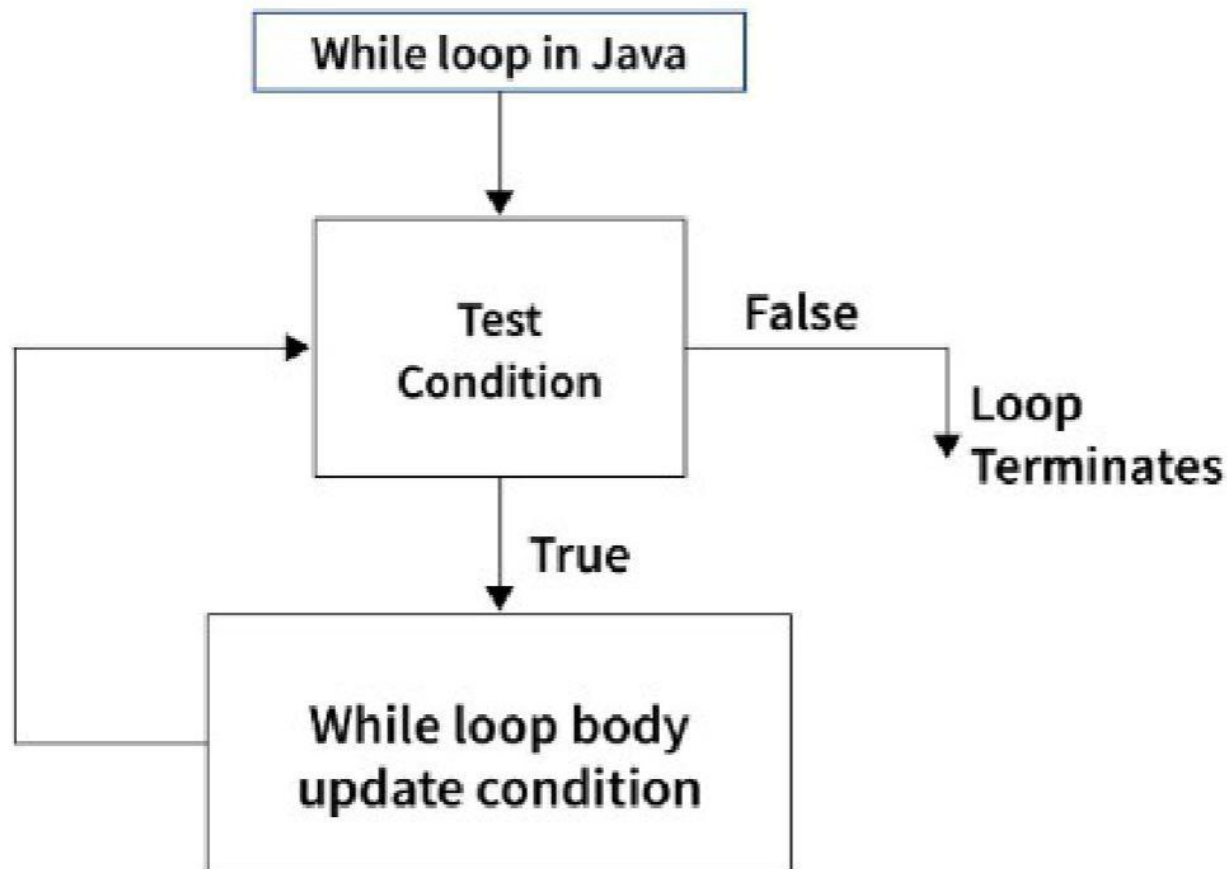
```
while(test_condition)
{
    //statements

    update_condition;
}
```

Repetition Statements

while loop

- Flowchart of While Loop in Java



Repetition Statements

while loop

- Example 1

```
public class Example {  
    public static void main(String[] args) {  
        int i=1;  
        while(i<6){  
            System.out.print(i+" ");  
            i*=2;  
        }  
    }  
}
```

Repetition Statements

while loop

- Example 1

```
public class Example {  
    public static void main(String[] args)  
    {  
        int i=1;  
        while(i<6) {  
            if(i==4) {  
                System.out.println(i);  
                break;  
            }  
            System.out.println(i);  
            i++;  
        }  
    }  
}
```

Repetition Statements

while loop

- **Class Work**

- 1 Write program that find the summation of numbers from 1 to 10.

Repetition Statements

while loop

- Answer for above question is

```
class whileLoopDemo {  
    public static void main(String args[])  
    {  
        int x = 1, sum = 0;  
        while (x <= 10) {  
            sum = sum + x;  
            x++;  
        }  
        System.out.println("Summation: " + sum);  
    }  
}
```

Repetition Statements

while loop

- **Class Work**

- 1 Write program that the Sum of Positive Numbers Only

Repetition Statements

while loop

- Answer for above question is

```
import java.util.Scanner;
class Student {
    public static void main(String[] args) {
        int sum = 0;
        Scanner input = new Scanner(System.in);
        System.out.println("Enter a number");
        int number = input.nextInt();
        while (number >= 0)
        {
            sum += number;
            System.out.println("Enter a number");
            number = input.nextInt();
        }
        System.out.println("Sum = " + sum);
        input.close();
    }
}
```

Repetition Statements

while loop

- **Class Work**

- 1 Write program that Finding factorial of a number entered by user using while loop

Repetition Statements

while loop

- Answer for above question is

```
import java.util.Scanner;
public class Example {
    public static void main(String[] args) {
        int number;
        System.out.println("Enter the number: ");
        Scanner scanner = new Scanner(System.in);
        number = scanner.nextInt();
        long fact = 1;
        int i = 1;
        while(i<=number){
            fact = fact * i;
            i++; }
        System.out.println("Factorial of "+number+" is: "+fact);}}
```

Repetition Statements

while loop

- **Home Work**

- 1 Write a java program to check whether a given number is Spy number or not.

Repetition Statements

do—while loop

- **do-while loop** is used to execute a block of statements **continuously until the given condition is true.**
- The do-while loop in Java is **similar to while loop**
 - ▶ except that the **condition is checked after the statements are executed**
- So **do while loop** guarantees the loop **execution at least once.**

Repetition Statements

do—while loop

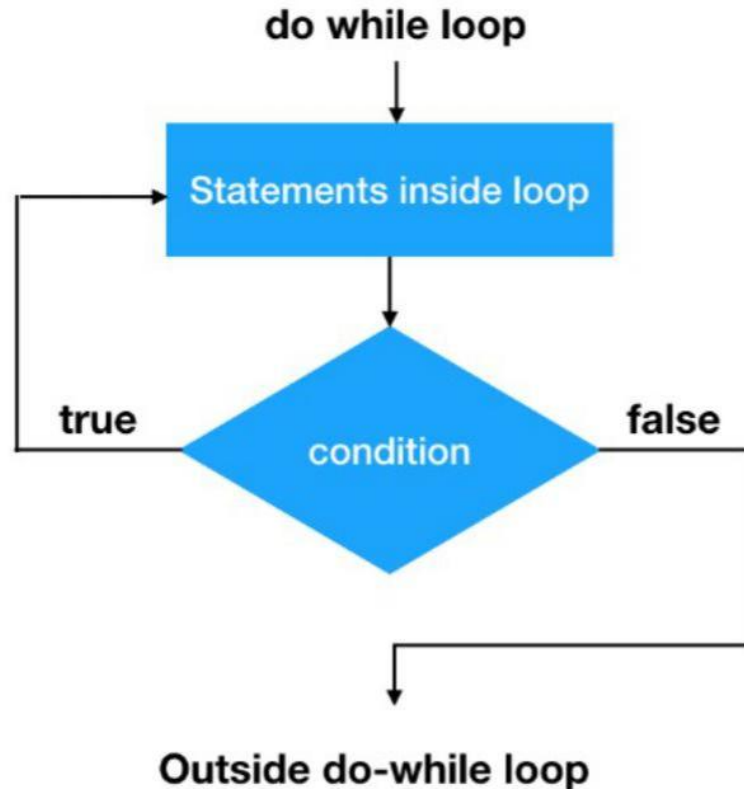
- do while loop syntax is as follows:

```
do {  
    // statements  
} while (expression);
```

Repetition Statements

do—while loop

- do while loop flow diagram



Repetition Statements

do—while loop

- Example1

```
public class JavaDoWhileLoop {  
    public static void main(String[] args) {  
  
        int i = 5;  
        do {  
            System.out.println(i);  
            i++;  
        } while (i <= 4);  
    }  
}
```

Repetition Statements

do—while loop

- Example2

```
class dowhileDemo
{
    public static void main(String args[])
    {
        int i=0;
        do
        {
            System.out.println(++i);
            ++i;
        }
        while(i<5);
    }
}
```

Repetition Statements

do—while loop

- Example3

```
class Example {  
    public static void main(String args[]){  
        int arr[]={2,11,45,9,65,45,46};  
        int i=1;  
        do{  
            System.out.println(arr[i/2]);  
            i++;  
        }  
        while(i<8);  
    }  
}
```

Repetition Statements

For loop

- **For Loop** in Java is a statement which allows code to be **repeatedly executed**.
- For loop contains **3 parts**
 - 1 Initialization
 - 2 Condition
 - 3 Increment or Decrements

for(a = 5; a <= 10; a++)

The diagram illustrates the three components of a for loop. The code `for( a = 5; a <= 10; a++ )` is shown in green. Red curly braces are placed under each part: `a = 5;`, `a <= 10;`, and `a++`. Blue vertical lines connect these braces to labels below: `Initialization` for the first part, `Condition` for the second, and `Increment (++) or Decrement (--)` for the third.

Initialization

Condition

Increment (++)
or
Decrement (--)

Repetition Statements

For loop

- Example

```
class ForLoopDemo
{
    public static void main(String args[])
    {
        for(int num = 1; num <= 5; num++)
        {
            System.out.println(num);
        }
        System.out.println("This for loop");
    }
}
```

Repetition Statements

For loop

- **Class Work**

- 1 Write a program to calculate the sum of first 10 natural number.
- 2 Write a program that prompts the user to input a positive integer. It should then print the multiplication table of that number.
- 3 Write a program to find the factorial value of any number entered through the keyboard.

Repetition Statements

For loop

- **Home Work**

- 1 Two numbers are entered through the keyboard. Write a program to find the value of one number raised to the power of another. (Do not use Java built-in method)
- 2 Write a program to print all the even numbers from 1 to 20 using for loop.

Repetition Statements

Nested Loop in Java

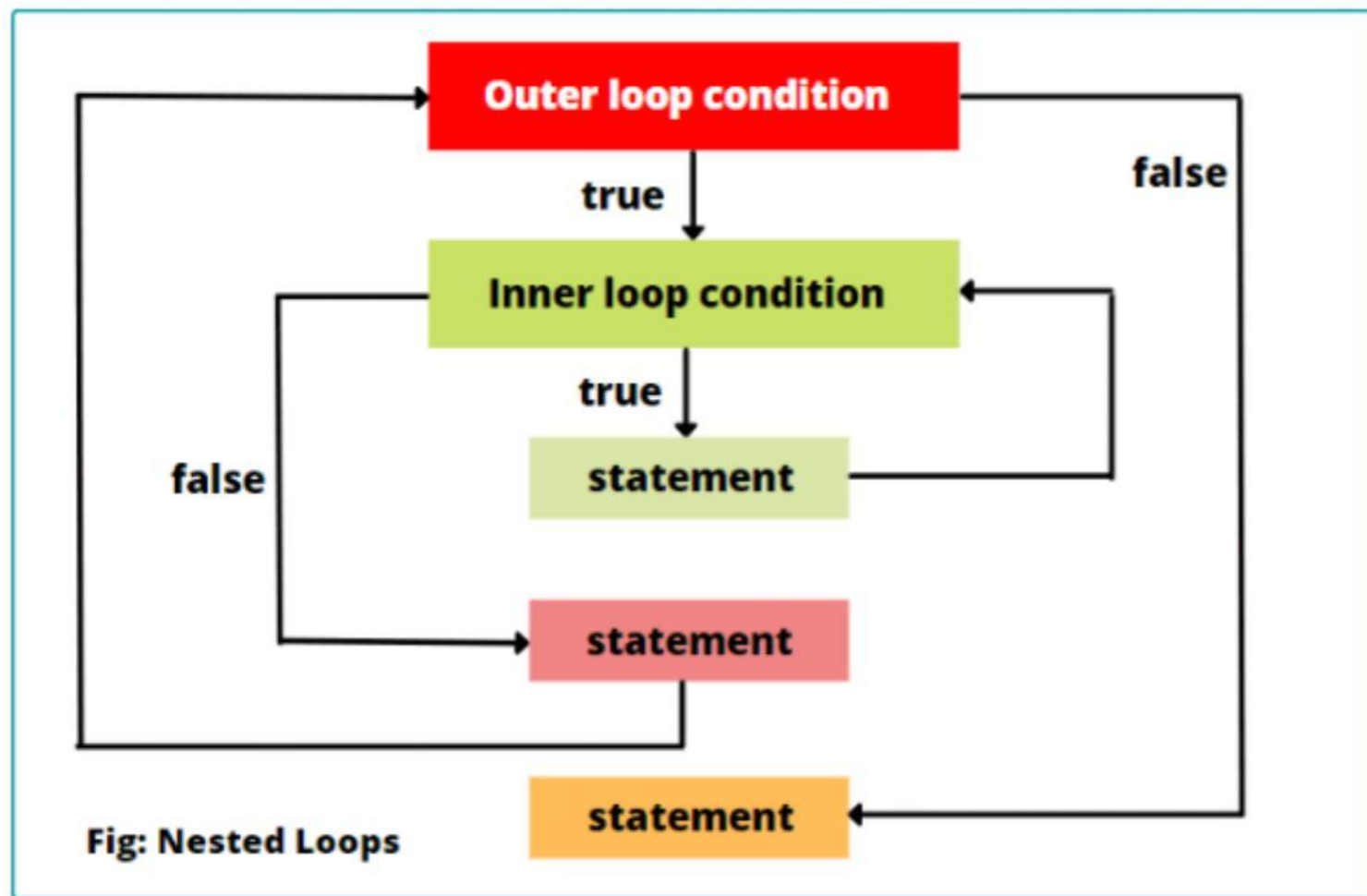
- **Nested loop** means a loop statement **inside another loop statement**.
- That is why nested loops are also called as “**loop inside loop**”.

```
for ( initialization; condition; increment )
{
    for ( initialization; condition; increment )
    {
        // statement of inside loop
    }

    // statement of outer loop
}
```

Repetition Statements

- Flowchart Diagram for Nested Loops in Java



Repetition Statements

Nested Loop in Java

- Examples

```
public class Nested
{
    public static void main(String[] args)
    {
        for(int i=1;i<=2;i++)
        {
            for(int j=1;j<=2;j++)
            {
                System.out.println(i+" "+j);
            }
        }
    }
}
```

Repetition Statements

Nested Loop in Java

- Examples

```
class Mainn
{
    public static void main(String args[] )
    {
        int i,j;
        for(i=1;i<=5;i++)
        {
            for(j=1;j<=i;j++)
            {
                System.out.print(i+"\t");
            }
        }
    }
}
```

Repetition Statements

Nested Loop in Java

- Examples 2

```
public class PatternEx {  
    public static void main(String[] args)  
    {  
        System.out.println("Displaying alphabet pattern: ");  
        for(int i = 65; i <= 69; i++)  
        {  
            for(int j = 65; j <= i; j++){  
                char ch = (char)j;  
                System.out.print(ch+ " ");  
            }  
            System.out.println(" ");  
        }  
    }  
}
```

Repetition Statements

Nested Loop in Java

- **class work 1**

- ▶ Write the Java program to display the following output



Repetition Statements

Nested Loop in Java

- **class work 2**

- ▶ Write the Java program to display the following output



Repetition Statements

Nested Loop in Java

- **class work 3**

- ▶ Write the Java program to display the following output

```
1
1 2 3 4
2
1 2 3 4
3
1 2 3 4
```

Repetition Statements

Nested Loop in Java

- Example

```
class Mainn
{
    public static void main(String[] args)
    {
        int x = 1;
        while(x < 2)
        {
            int y = 1;
            while(y < 4)
            {
                System.out.println(x + ":" + y);
                y++;
            }
            x++;
        }
    }
}
```

Repetition Statements

Nested Loop in Java

- Example

```
class NestedWhileExample
{
    public static void main(String arg[])
    {
        int a = 1;
        while(a < 3)
        {
            int b = 5;
            while(b < 8)
            {
                System.out.println(a + " " + b);
                b++;
            }
            a++;
        }
    }
}
```

Break and Continue statement in Java

Break statement in Java

- The **break statement** is usually used in following **two scenarios**:
 - ① Use break statement to come out of the **loop instantly**.
 - ★ Whenever a break statement is encountered inside a loop, the control directly comes out of loop and the **loop gets terminated for rest of the iterations**.
 - ★ It is used along with if statement, whenever used inside loop so that the **loop gets terminated for a particular condition**.
 - ★ The important point to note here is that when a **break statement** is used **inside a nested loop**, then **only the inner loop gets terminated**.
 - ② It is also used in **switch case control**.

Syntax :

► **break;**

Break and Continue statement in Java

Break Statement in Java

- Examples 1

```
public class Sub {  
    public static void main(String args[]){  
        int num =0;  
        while(num<=10){  
            System.out.println("Value of variable is: "+num);  
            if (num==2)  
            {  
                break;  
            }  
            num++;  
        }  
        System.out.println("Out of while-loop");}}}
```

Break and Continue statement in Java

Break Statement in Java

- Output of the above example is

```
Value of variable is: 0  
Value of variable is: 1  
Value of variable is: 2  
Out of while-loop
```

Break and Continue statement in Java

Break Statement in Java

- Examples 2

```
public class Sub {  
    public static void main(String args[]){  
        int var;  
        for (var =100; var>=10; var --)  
        {  
            System.out.println("var: "+var);  
            if (var==99){  
                System.out.println("var: "+var);  
                break;  
            }  
        }  
        System.out.println("Out of for-loop");}}
```

Break and Continue statement in Java

Break Statement in Java

- Examples 3

```
public class Sub {  
    public static void main(String[] args) {  
        int i=12;  
        do {  
            if(i==15){  
                ++i;  
                int y=5;  
                y++;  
                ++y;  
                i=y++;  
                System.out.println(i);  
                break;}  
            System.out.println(i);  
            i++;}  
        while(i<=20);}}
```

Break and Continue statement in Java

Break Statement in Java

- Examples 4

```
public class Mainn{
    public static void main(String[] args)
    {
        for(int i=1;i<=2;i++)
        {
            for (int j = 0; j <=3; j++)
            {
                if(j==2)
                    break;
                System.out.print(j);
            }
        }
    }
}
```

Break and Continue statement in Java

Using break as a Form of Goto

- Java **does not have a goto statement**
- Now, the break statements can be used to **jump out of the target block.**

Break and Continue statement in Java

Using break as a Form of Goto

- Example

```
class GFG {  
    public static void main(String args[])  
    {  
        first:  
        for (int i = 0; i < 3; i++)  
        {  
            second:  
            for (int j = 0; j < 3; j++)  
            {  
                if (i == 1 && j == 1)  
                {  
                    break first;  
                }  
                System.out.println(i + " " + j);  
            }  
        }  
    }  
}
```

Break and Continue statement in Java

Using break as a Form of Goto

- Example

```
class GFG {  
    public static void main(String args[])  
    {  
        first:  
        for (int i = 0; i < 3; i++)  
        {  
            second:  
            for (int j = 0; j < 3; j++)  
            {  
                if (i == 1 && j == 1)  
                {  
                    break second;  
                }  
                System.out.println(i + " " + j);  
            }  
        }  
    }  
}
```

Break and Continue statement in Java

Using break as a Form of Goto

- Example

```
class Test
{
    public static void main(String[] args)
    {
        out:
        for(int i=1; i<=100; i++)
        {
            System.out.println("outer");
            for(int j=1; j<=100; j++)
            {
                System.out.println("nested");
                if(j==2)
                {
                    break out;
                }
            }
        }
    }
}
```

Break and Continue statement in Java

Using break as a Form of Goto

- Example

```
public class Sub {  
    public static void main(String[] args) {  
        outer:  
        for (int i = 0; i < 10; i++) {  
            for (int j = 0; j < 10; j++) {  
                if (j == 1)  
                    break outer;  
                System.out.println(" value of j = " + j);  
            }  
        }  
    }  
}
```

Break and Continue statement in Java

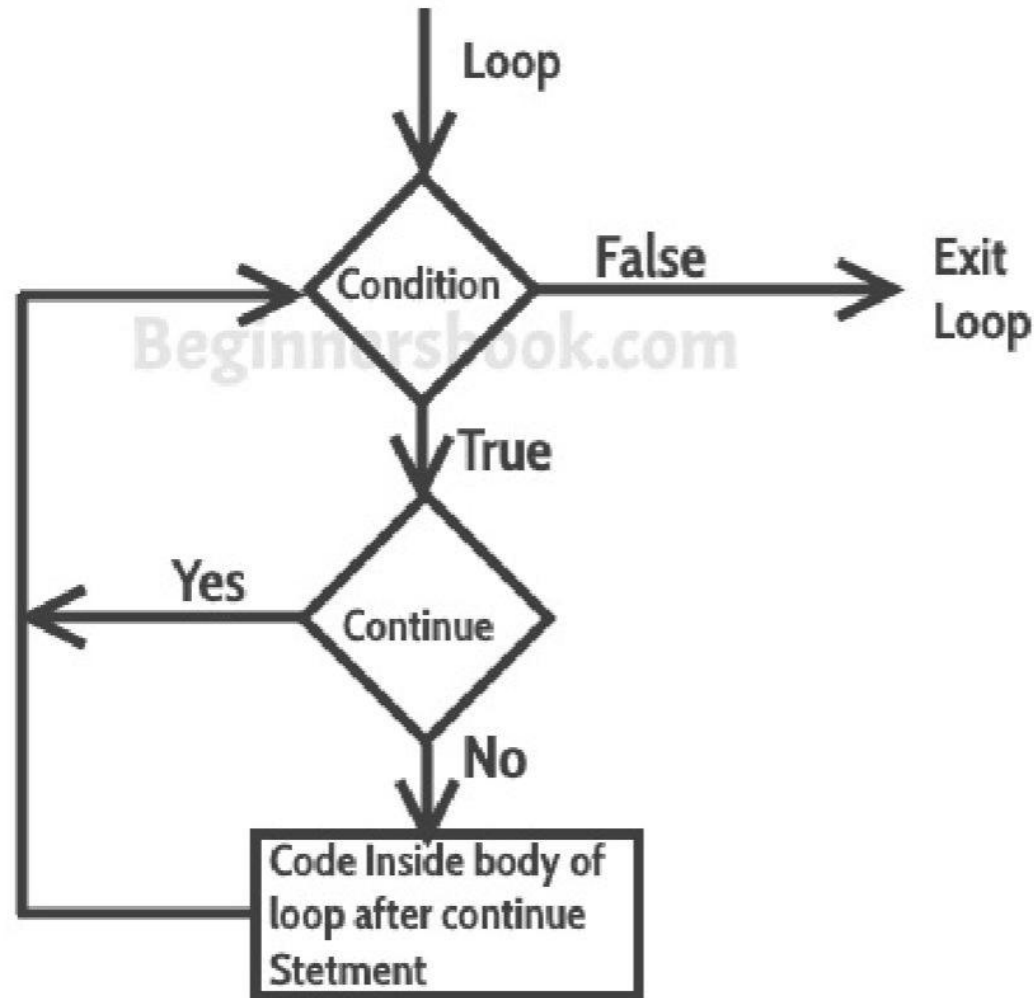
continue Statement

- In Java, the continue statement is used to **skip the current iteration** of the loop.
- It jumps to the **next iteration** of the loop **immediately**.

Break and Continue statement in Java

continue Statement

- Flow chart of continue Statement



Break and Continue statement in Java

continue Statement

- Example

```
public class ContinueDemo1
{
    public static void main(String[] args)
    {
        for(int i=1; i<=10; i++)
        {
            if(i==5)
            {
                continue;
            }
            System.out.println(i);
        }
    }
}
```

Break and Continue statement in Java

continue Statement

- Example2

```
public class ContinueDemo2 {  
    public static void main(String[] args)  
    xy:  
        for(int i=1;i<=3;i++){  
            pq:  
                for(int j=1;j<=2;j++){  
                    if(i==2&& j==2)  
                    {  
                        continue xy;  
                    }  
                    System.out.println(i+" "+j);  
                }  
            }  
        }  
    }  
}
```

Break and Continue statement in Java

continue Statement

- Example2

```
public class ContinueDemo2
{
    public static void main(String[] args)
    {
        xy:
        for(int i=1;i<=3;i++){
            pq:
            for(int j=1;j<=2;j++)
            {
                if(i==2 || j==2)
                {
                    System.out.println(i+" "+j);
                    continue xy;
                }
                System.out.println(i+" "+j);
            }
        }
    }
}
```

Break and Continue statement in Java

continue Statement

- Example3

```
public class ContinueDemo2 {  
    public static void main(String[] args) {  
        xy:  
        for(int i=1;i<=3;i++){  
            pq:  
            for(int j=1;j<=2;j++){  
                {  
                    if(i==2&& j==2)  
                    {  
                        continue pq;  
                    }  
                }  
                System.out.println(i+" "+j);  
            }  
        }  
    }  
}
```

Break and Continue statement in Java

continue Statement

- Example4

```
class Main {  
    public static void main(String[] args) {  
        first:  
        for (int i = 1; i < 6; ++i)  
        {  
            for (int j = 1; j < 5; ++j)  
            {  
                if (i == 3 || j == 2)  
                    continue first;  
                System.out.println("i = " + i + "; j = " + j);  
            }  
        }  
    }  
}
```

Break and Continue statement in Java

continue Statement

- Example5

```
public class Mainn {  
    public static void main(String[] args){  
        int num = 1;  
        while(num<5){  
            if (num%3==0){  
                num++;  
                continue;  
            }  
            System.out.println("counting...."+num);  
            num++;  
        }  
    }  
}
```

Revision Question

Increment and Decrement

- Example

```
public class IncrementDecrement
{
    public static void main(String[] args)
    {
        int number = 50;
        System.out.println("Number is " + number);
        number++;
        System.out.println("Now, number is " + number);
        number--;
        System.out.println("Now, number is " + number);
    }
}
```

Revision Question

Increment and Decrement

- Example2

```
public class Mainn
{
    public static void main(String[] args)
    {
        int number = 50;
        int x;
        System.out.println("Number is " + number);
        x=number++;
        System.out.println("Now, number is " + number);
        System.out.println("Now, x is " + x);
        x=number--;
        System.out.println("Now, number is " + number);
        System.out.println("Now, x is " + x);
    }
}
```

Revision Question

Increment and Decrement

- Example3

```
public class JavaProgram
{
    public static void main(String args[])
    {
        int a = 1;
        int b = 2;
        int c;
        int d;
        c = ++b;
        d = a++;
        c++;
        System.out.println("a = " +a);
        System.out.println("b = " +b);
        System.out.println("c = " +c);
        System.out.println("d = " +d);}}}
```

Revision Question

Increment and Decrement

- Example4

```
public class JavaProgram
{
    public static void main(String args[])
    {

        int a = 1;
        int b = 2;
        int c;
        int d;
        c = --b;
        d = a--;
        c--;
        System.out.println("a = " +a);
        System.out.println("b = " +b);
        System.out.println("c = " +c);
        System.out.println("d = " +d);}}
```

Revision Question

Increment and Decrement

- Example5

```
class Mainn
{
    public static void main(String[] args)
    {
        int x = 1;
        while(x < 2)
        {
            int y = 1;
            while(y < 4)
            {
                System.out.println(x + ":" + y);
                y++;
            }
            x++;
        }
    }
}
```

Revision Question

Increment and Decrement

- Example6

```
class NestedWhileExample
{
    public static void main(String arg[])
    {
        int a = 1;
        while(a < 3)
        {
            int b = 5;
            while(b < 8)
            {
                System.out.println(a + " " + b);
                b++;
            }
            a++;
        }
    }
}
```

Revision Question

Increment and Decrement

- Example6

```
public class Driver1 {  
    public static void main(String[] args) {  
        System.out.println("Test Break statement in While loop");  
        int i = 0;  
        while (i < 5) {  
            if (i == 2) {  
                break;  
            }  
            System.out.println(i++);  
        }  
    }  
}
```

Chapter Three

- **End of Chapter Three**